AI Mechanistic Interpretability and Self-Aware Networks:

From Neural Decoding to Causal Self-Regulation

Author: Micah Blumberg

Self Aware Networks Research Institute

https://selfawarenetworks.com

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Abstract

An internal representation can be decodable, causally effective and numerically reconstructable without implementing the operation a user intended. I examine these distinctions through the receiver-relative framework of Self-Aware Networks, connecting a source-recovered neural-decoding and feedback question to explicit transformer interventions. The research combines intervention prediction in controlled learned networks, passive and active feedback controls, GPT-2 activation and cache experiments, and native hybrid-state interventions in a pinned quantized Qwen3.5-0.8B decoder. Complete corrected-state transfer supports all eight tested bidirectional story groups in an initial Qwen assay, while attention-only transfer fails one; a complementary intervention gives identical attention state with different behavior. In a separate learned-correction assay, linear and nonlinear attention-state predictors each obtain five of 32 requested reversed-role answers but no story with both roles correctly reversed. Both preserve the tested color attribute. Lower training loss and partial-state error therefore fail to certify the intended relational change. Formal arguments distinguish observational from causal identification, partial-state from behavioral sufficiency, and output-margin guarantees from reconstruction loss. Earlier failures, nulls and inadequate baselines are retained. The contribution is a bounded experimental account of receiver-dependent intervention and its certification problem, not invention of mechanistic interpretability, evidence of phenomenal consciousness, or demonstrated autonomous self-regulation in frontier models.

Keywords: AI mechanistic interpretability; Self-Aware Networks; causal intervention; neurofeedback; hybrid state; representation learning; GPT-2; Qwen; semantic preservation.

1 Interpreting a representation requires its receiving computation

Suppose a model's internal state can be decoded as representing a person, object or relation. That is a useful observation, but several different questions remain. Does the identified state participate in the answer? Does changing it implement the requested operation? Does it retain that effect when the question changes? Can feedback help the system learn a useful correction? And can an apparently successful change be reversed without disturbing other retained computation?

My receiver-relative SAN account treats these as questions about one connected process. A signal does not arrive with its complete consequence attached. The receiving network's learned organization, current state and active routes determine how it is transformed and what follows. The artificial translation must therefore inspect downstream computation, rather than stopping at a readable feature label. A decoded "giver" feature, for example, must support the appropriate relation between giver and recipient, not merely make one person's name more likely in every answer.

This framing preserves the original ambition of Neo Mind Cycle: measured internal activity might become useful returned information for a learning system. It does not assume that a model already possesses an endogenous instrument for reading its own activations. An experimenter can build a measurement-and-return loop around a system; a different experiment is needed to determine what the system itself can observe, control and learn. A reported intention, a changed probe score and an independently useful action are not interchangeable outcomes.

The present paper develops that question through a sequence of explicitly bounded experiments. Its central modern-model contribution is a native hybrid-state study and a subsequent attempt to learn the correction without access to a corrected test state. The favorable state-transfer result is preserved alongside the failed learned operation. The companion *AI Core Alignment Across Scales: Human-Guided Co-Regulation in Self-Aware Artificial Networks* uses shared data to examine authorization and recovery. Those data are not independent replications, and this paper does not own the companion's authority architecture.

Neither paper identifies token position with neural phase, residual vectors with brainwaves, attention weights with cortical connections, or expert routing with a thalamus. A proposed functional correspondence needs a concrete implementation on each substrate. The phrase *Self-Aware Networks* names the motivating theory; it does not make the present model experiments evidence of subjective experience.

2 The source-recovered question

My May 2012 note distinguishes decoding brain signals into commands from transforming measured signals into light or sound that the receiving brain could learn to use [1]. A neighboring retrospective source acknowledges uncertainty about how that interpretation would work [2]. These qualifications belong in the genealogy.

The 2017 Android Jones/Neural Lace article develops a closed-loop idea: measure the current pattern and return a difference relative to a desired pattern [3]. A later transcript asks how to determine that an optimization was genuinely good or better, rather than merely changed [4]. Its machine transcript is not an audio-verified quotation, and its assistant-written header is not historical author testimony.

In a February 2025 working document, I explicitly named a proposed "Bio & Mechanistic Interpretability" project [5]. An Oscillatory Cortical Algorithm draft discusses biological decoding alongside artificial-network interpretation [6]. The first occurrence recovered in local Git of the explicit 2025 title has a February 17, 2025 commit timestamp. This is not independently verified public custody.

The joined question is substantial but bounded: can measurements become usable feedback through a learned receiving system, and can the resulting loop be evaluated causally and behaviorally? Existing neural decoding, neurofeedback, and mechanistic interpretability predates or independently develops parts of this question. The paper does not claim their invention.

The point of recovering this context is not to substitute an early date for experimental evidence. It is to preserve the full operation being proposed: measurement, a receiving system capable of learning its use, returned consequences and an assessment of whether the change helped. Reducing that proposal to "the brain is a hard drive" or "a probe reveals a mind" would discard the very learning and evaluation problem that requires investigation.

3 Existing achievements and the remaining distinction

Kay and colleagues demonstrate a substantive neural-decoding achievement: identifying viewed natural images from measured human brain activity [7]. Decoding is not thereby a demonstration of writing those experiences into the brain. The same distinction matters for artificial systems. Causal abstraction formalizes relationships between computational levels [8], and circuit tracing uses interventions and computational graphs rather than relying only on verbal labels [9]. These are strong existing methodologies, not passive baselines to be dismissed.

Gurnee and colleagues study representations available for verbal report, modulation and flexible downstream use [10]. Their functional workspace question is distinct from a claim about subjective experience. Ji-An and colleagues use in-context neurofeedback to investigate reporting and control of activation patterns, with dependence on the chosen direction and examples [11]. Aoki and colleagues pose a more restrictive privileged-access test and report no reliable control under that setting [12]. Their difference motivates controlling what the model and an external observer can infer; it does not establish that one study tested every interpretation of the other.

ReFT already learns interventions on frozen language-model representations [18]. RAVEL explicitly evaluates intended changes, isolation of unrelated attributes and generalization [17]. The experiments here are not full reproductions or superiority tests of those methods. Their residual contribution is the conjunction of native hybrid-state support, a query-independent learned correction, whole generated answers, relational and collateral endpoints, and explicit authorization/restoration boundaries. Familiar algebra or a new collection of plots cannot establish novelty by itself.

Within the SAN corpus, *The External Cortex Loop* concerns a human measurement-and-feedback protocol, *The Network That Learns It Is a Network* concerns explicit network self-knowledge, and *Reciprocal AI-Neuroscience Translation* develops the broader translation method. Their full-text ownership comparison is retained in the companion source record. The current object is narrower: whether a proposed explanation or correction predicts the intended operation in the original artificial receiving computation.

4 Mathematical distinctions and experimental obligations

4.1 Decoding, causal use and prediction

Let h denote an internal state, c the receiving context and f a specified scalar readout of the original downstream computation. A decoder D and an intervention δ answer different questions:

$$\widehat{z} = D(h), \qquad \Delta(h, c, \delta) = f(h + \delta, c) - f(h, c). \tag{1}$$

The first expression predicts a label. The second measures an effect in the receiving model. A readable label does not prove that its putative feature caused an answer, and a nonzero effect does not prove the intended operation. Strong interventions may damage unrelated computation or move the state outside its naturally encountered distribution.

An explanatory surrogate $\widehat{\Delta}$ should predict held-out original-model effects over a declared test set $\mathcal{T}$. One elementary evaluation is

$$E_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{(h,c,\delta)\in\mathcal{T}} |\widehat{\Delta}(h,c,\delta) - \Delta(h,c,\delta)|. \tag{2}$$

The locality of an approximation matters. Differentiability gives the familiar first-order relation

$$f(h+\delta, c) - f(h,c) \approx J_f(h,c)\delta. \tag{3}$$

A context-specific Jacobian has access to the original downstream mechanism at the evaluated state. Its advantage over an amortized decoder is not free. Conversely, a context-averaged or wrong-context Jacobian need not predict the same intervention. These are established chain-rule consequences, not newly discovered SAN mathematics.

4.2 A feedback loop need not identify endogenous control

Consider two scalar systems starting from $h_0 = 1/2$, with common update $h_{t+1} = 0.6 + 0.3q_t$. Their feedback policies differ:

$$q_t^+ = h_t, \qquad q_t^- = 1 - h_t. \tag{4}$$

The first converges to $6/7$, the second to $9/13$; their limiting contrast is $15/91$. This follows by solving each fixed-point equation. Errors multiply by 0.3 and -0.3 respectively. The construction can produce a feedback-associated difference without any adaptive internal policy choosing an action. It shows why context-mediated dynamics alone do not establish endogenous self-regulation.

An active learner can add an actual update. For example, the reference bandit uses

$$Q_{t+1}(a_t) = \tfrac{3}{4}Q_t(a_t) + \tfrac{1}{4}r_t, \tag{5}$$

with other action values retained. This is standard incremental learning [16]. Whether the learner is "internal" is not identified merely by attaching that label. An external controller with the same observations, state update and action interface can implement the same policy. A claimed endogenous advantage requires a difference in accessible information or mechanism, measured under appropriate controls.

4.3 Operation-level and state-level certificates

Suppose an intended role transformation must work in both directions. A constant answer (a, a) can score one of two when targets differ, but cannot implement both required mappings. Aggregation over individual answers can therefore conceal relation collapse. For two directions and three required queries, define a complete group:

$$G_g = \prod_{d\in\{\rightarrow,\leftarrow\}} \prod_{q\in\{\text{giver},\text{recipient},\text{color}\}} \mathbf{1}[\widehat{a}_{g,d,q} = a^*_{g,d,q}]. \tag{6}$$

This criterion is task-specific, not a complete account of meaning. Its advantage is that it states the requested operation before accepting a favorable average. It also requires baseline competence, since failure of an unedited model to answer the underlying questions can make a joint zero non-diagnostic.

Let s be complete model state, $\pi(s)$ a retained projection such as the attention cache, and $F(s, q)$ the full generated answer. Behavioral sufficiency of that projection requires

$$\pi(s) = \pi(t) \Longrightarrow F(s, q) = F(t, q) \quad \text{for all declared } s, t, q. \tag{7}$$

On the declared domain this is equivalent to factorization through π: define the reduced function from any representative of each fiber; Equation (7) makes the definition independent of that choice. A single equal-projection/different-answer pair disproves this sufficiency on a domain containing the pair. It does not show that the projection is uninformative, or that no narrower task domain could factor through it.

5 The evidence that shaped the native-model test

5.1 Controlled learned networks and a preserved leakage repair

The first intervention-prediction application trains small networks with hidden layers of 24 and 16 units on designed world states. Its development classifiers achieve balanced accuracy near 0.90, while linear probes are less accurate. A local contextual Jacobian predicts intervention effects more accurately than several fitted surrogates. However, 60 selected donor states fall outside training, including 32 test states. The resulting holdout interpretation is invalid. Those outputs remain recorded as contaminated development, not confirmation.

The repaired split closes all donor-related variants within 64 families: 32 training, 16 development and 16 test. Three seeds yield 18,432 effect rows and 72 metric rows. Test balanced accuracy lies approximately between 0.72 and 0.78. Mean effect errors are 0.3395 for linear, 0.3311 for polynomial, 0.3367 for unit/context surrogates and 0.076 for local contextual Jacobians. Mean and wrong-context Jacobians give 0.3383 and 0.4729. The useful result concerns receiver-specific prediction in these networks. Local derivative access, differing computational cost and the few underlying networks prevent treating the row count as independent replications or a general model-size effect.

5.2 Passive versus active returned information

A MiniLM sentence-encoder assay collects 88 measured states: 11 displayed score prefixes for each of eight sentences [13]. Its probe is fitted on 16 synthetic ethics-labelled examples. Perfect training fit is not ethical calibration. Six passive-feedback arms over 20 iterations produce 960 lookup steps from those measured states. Three sentences show positive contrast and five show none; the mean contrast is 0.002819. Across each sentence's full displayed-score grid, the range is at least 0.024. Thus context can change the measured representation, but this particular passive loop does not demonstrate useful learned self-regulation. There is no learned action policy or autonomous choice of response in that assay.

An active contextual-bandit application then supplies a real learning update, three actions, opaque grant wires and a two-phase drift. With 48 feedback observations per context and phase, internal and task-feedback conditions both restore the target route relation and achieve 26 of 48 useful decisions. An equal-information external controller matches them. At the smaller 12-observation budget, internal and task-feedback conditions differ, but their feedback channels carry unequal information. Across 84 conditions the application retains 7,200 feedback events and 33,792 evaluation decisions. The result supports an instrument-assisted learning loop, not a privileged internal faculty. Real forged/replayed feedback tests subsequently demonstrate that an unprotected update channel can damage that learner.

These distinctions answer the historical question about “good or better” operationally. A changed measured score can be real without establishing useful behavior; a useful policy can be learned without proving endogenous access. Their separation is central to interpreting neurofeedback, not a reason to erase favorable or adverse results.

5.3 GPT-2: causal effect, learned correction and a baseline floor

The pinned quantized GPT-2 small graph supplies actual causal attention, value/output routes, residual updates, feed-forward computation and retained keys/values [14,19,21]. The static assay obtains 240 vocabulary readouts from 48 prompts. A selected whole-residual patch restores seven of 16 outputs, with 11 clean and zero corrupted outputs. Mean donor-versus-other logit contrast moves from -2.6633 to 1.3495, compared with 2.6633 clean. Matched norm and wrong-location controls do not restore the selected outputs. A contrast restricted to two names is not full-vocabulary answer accuracy, and this coarse-hook assay is not Wang and colleagues’ complete circuit replication [20].

The next assay learns a rank-four correction in a fixed projected representation. It fits 20 coefficients and retains 3,845 preprocessing values, using eight training cases and 16 new test cases. Internal feedback produces four correct outputs, visible task feedback five, yoked feedback four and supervised ridge nine; clean and privileged donor conditions produce 14 and 12. The learned internal-feedback condition passes zero of eight bidirectional pairs. Its positive individual answers do not establish the relational operation. Actual signed feedback, poisoned inputs and revocation are separately tested; an authorized update is not automatically a useful update.

A prospective structured-intent study adds 64 cases, 512 actual readouts and 1,536 reused outcome scores, with keep/swap instructions, two role queries and color checks. All learned conditions fail the full eight-outcome group criterion. Privileged donor and corrected-text conditions pass two and eight of 16 groups respectively. A reflection control has double-application error at most 1.91×10^{-6}, yet does not preserve the full semantic endpoint. Different query-terminal edits in this study are not one common edited world state.

The algebra explains the limit of that reflection result. For orthogonal projection P,

$$T(h) = \mu + (I - 2P)(h - \mu), \qquad T(T(h)) = h. \tag{8}$$

The identity follows from $P^2 = P$. It preserves the orthogonal complement and is reversible, but supplies no premise relating P to the desired role transformation. Norm clipping can also break the identity. Numerical reversibility therefore neither proves nor substitutes for semantic success.

A further GPT-2 continuation study retains the edited cache across later fixed inputs. It performs 884 readouts, including 16 fresh answers after restoration, and separates lower attention routes computed before the selected residual edit from later affected routes. However, the natural-language giver probe has zero of 32 correct ordinary answers. Its joint semantic zeros are consequently non-evaluable as evidence of correction failure. Cache identity and logit changes remain measurable. The next model/task pair was chosen to establish ordinary task competence before interpreting relational outcomes.

6 Mapping and intervening in a native hybrid decoder

6.1 Exact model and state

The contemporary application executes the text-only decoder from Qwen3.5-0.8B, using pinned upstream and quantized ONNX-conversion revisions [22,23]. The 24-layer graph combines 18 gated-delta layers with six full-attention layers. It has hidden width 1,024, 18 convolution-state tensors, 18 recurrent-state tensors and 12 key/value tensors. The retained states are part of the original executed graph, not a learned surrogate of its output. Gated-delta recurrence is established architecture [24]; the related DeltaNet-2 formulation [25] is not substituted for the executed model.

The hybrid distinction matters operationally. Processing a prefix produces state $s = (C, R, K)$. Each continuation uses a copy of that state and fresh query tokens, with a fixed non-thinking chat template, whole-vocabulary greedy decoding, both configured termination tokens and a four-token answer ceiling. The generated answer is not inserted into a different query branch. Exact model, tokenizer and runtime identities are supplied in Appendix A.

Figure 1 distinguishes complete donor transfer from selective transfer. A corrected-prefix donor is privileged test information. A successful interchange shows that the supplied state can change behavior; it does not show that the recipient learned to construct that state from a correction.

6.2 Development failure and the frozen transfer result

An initial development gate requires exact-case answers and fails at eight of 12 because four colors use different capitalization. The post-hoc case-folded result is 12 of 12. Both results are retained. The subsequent new-family protocol specifies trimming and case folding prospectively, records exact-case answers separately and includes every fixed case. Its ordinary baseline is 48 of 48.

The six conditions are original, complete donor, convolution donor, recurrent donor, attention donor and combined convolution/recurrent donor. Complete transfer reverses all 32 required role answers and preserves all 16 colors. Attention-only transfer reverses 31 role answers and all colors, failing one of eight complete groups. The other three selective conditions reverse zero, two and one role answers, respectively. Figure 2 shows this asymmetry without treating differences in group dimensionality or computational role as a universal importance ranking.

The individual exceptions carry more information than an average. In the garden story where Liam gives Emma a green book, attention-only donor transfer answers Emma to both role queries. In the opposite-direction garden story, changing the convolution/recurrent groups while retaining the original attention state changes the recipient answer. The latter supplies equal attention state with unequal behavior, refuting Equation (7) for the attention projection on the tested mixed-state domain. It does not establish that all natural states require all groups for every query.

The assay contains 288 primary answers and 12 fresh workflow answers, 4,608 state-origin records and 776 native decoder calls. All 300 answers and 32 prefix outputs are reconstructed through separate code. The mixed states may be off-manifold; full-state donor access is privileged; the four family combinations are small. These limits remain part of the interpretation even though the native readback is exact.

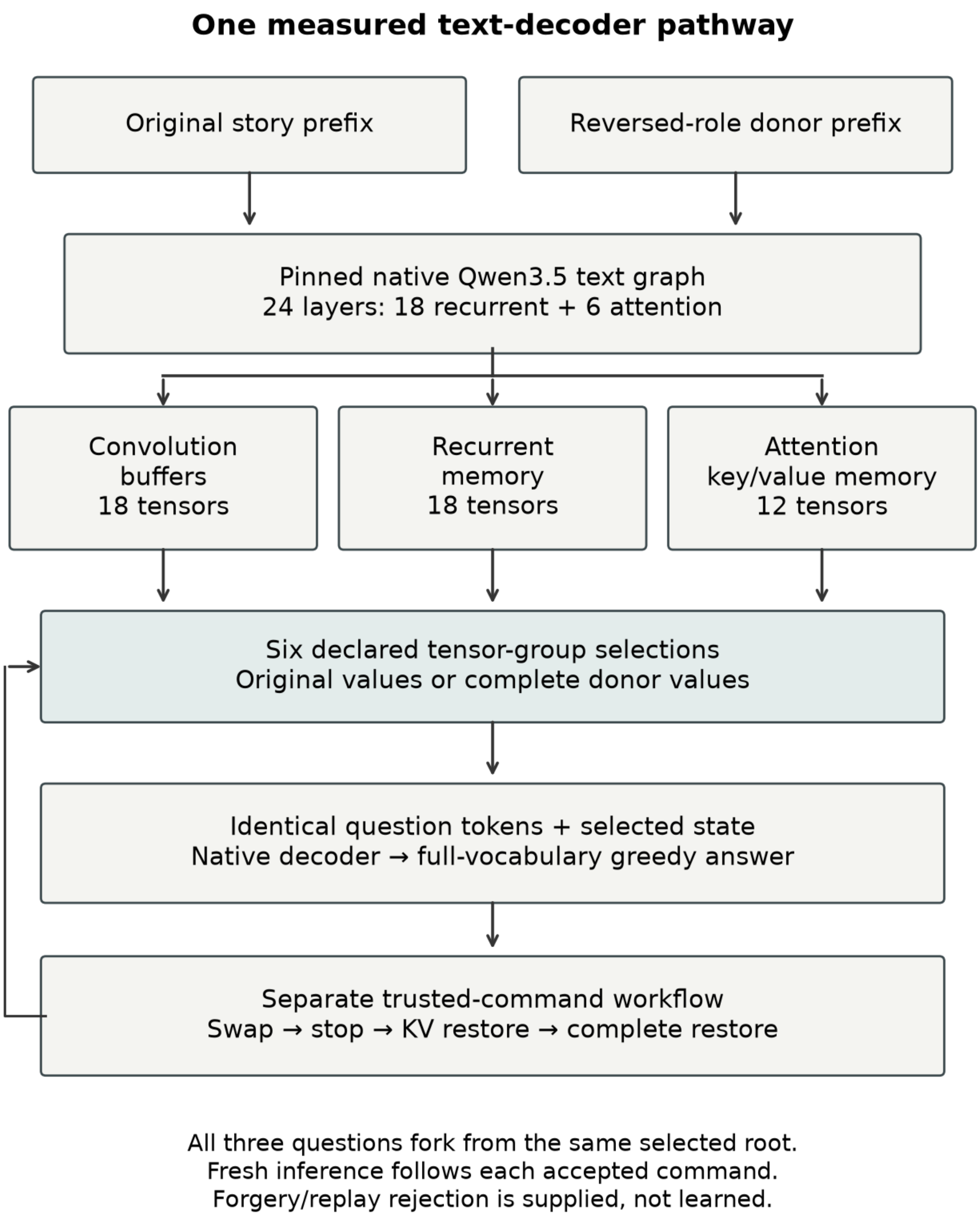

Figure 1: State and query boundaries in the native Qwen hybrid application. A single modified prefix is branched into independent queries. Selective transplantation retains the unselected original state groups; complete transplantation supplies the counterfactual groups together.

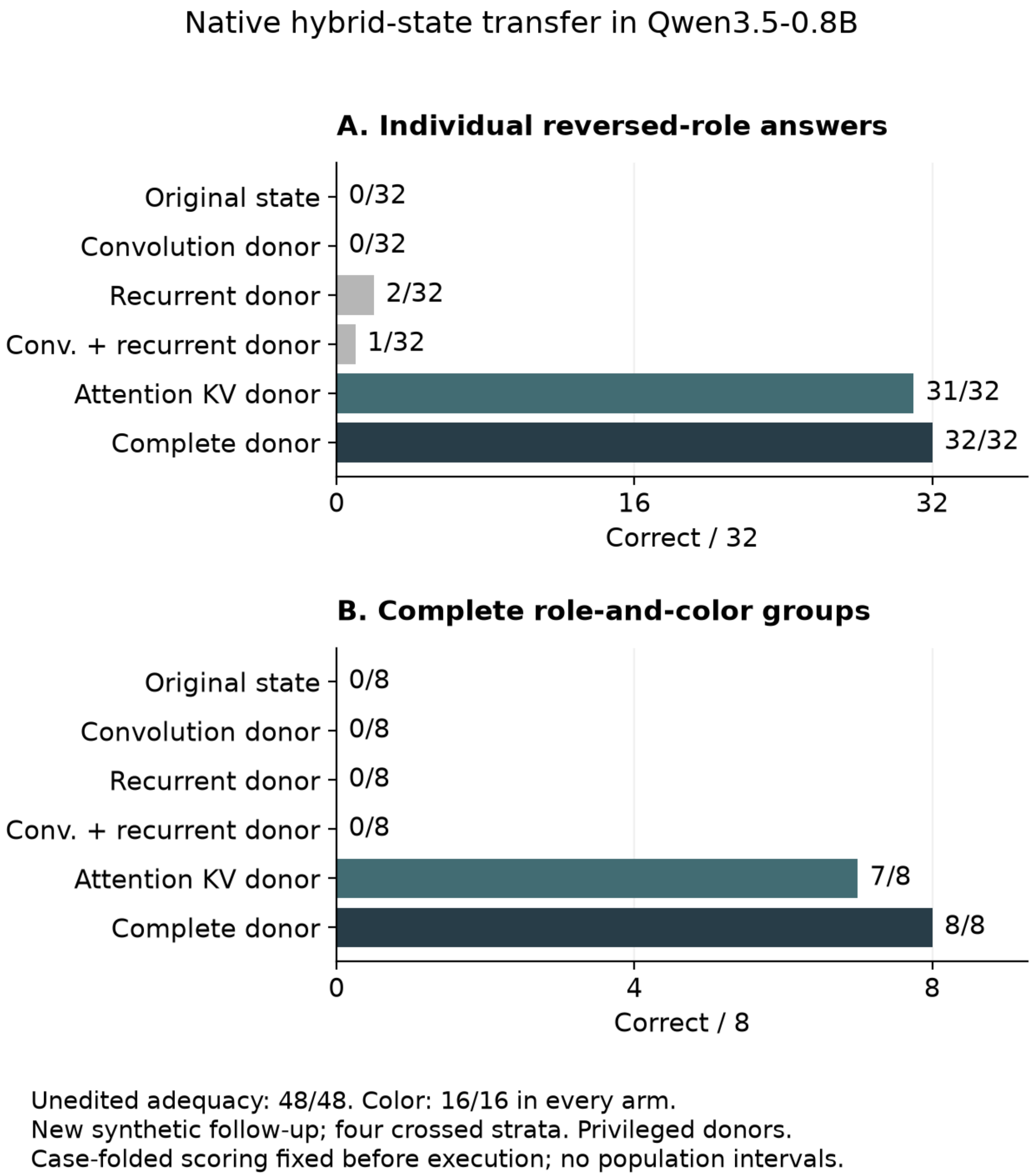

Figure 2: Native hybrid-state transfer outcomes. Complete donor transfer succeeds in all eight direction-paired groups; attention-only transfer fails one. All conditions preserve the tested color answers. These are counterfactual-state controls, not learned corrections.

6.3 Continuation, stopping and restoration

For a deterministic decoder with fixed computation, inputs and settings, equality of complete initial state implies equality of subsequent states and outputs by induction over the same tokens. This does not follow from equality of one cache projection. It also does not follow merely because the first answer happens to match.

The application distinguishes disabling future edits, restoring attention only and restoring a saved complete hybrid state. Disabling an edit leaves its past effect in place while normal inference continues. Attention-only restoration may recover an answer without recovering full logits or all retained state. Complete restoration requires the corresponding saved original state. None of these operations can undo a prior external action. These distinctions make a purportedly controllable intervention auditable rather than relying on an ambiguous "off" switch.

7 Learning the correction without a test donor

7.1 Dataset, fitting and intervention support

The learned study freezes 24 training prefixes and 16 new test prefixes before fitting. Six training name/context families include both role directions and two colors. Test names, contexts and colors are absent from the fit. One test context retains the training construction, while another jointly changes setting and verb. The small crossed design is not a population sample or a separately identified syntax-generalization effect.

Each representation is a nine-token window from the 12 attention tensors, with total dimension $D = 55{,}296$. Positions come from the controlled grammar. Let z_i be training-only centered and tensor-scaled states, and y_i the correspondingly scaled paired target changes. The two kernels and common dual fit are

$$\begin{aligned} K_{ij}^{\mathrm{lin}} &= z_i^\top z_j / D, \\ K_{ij}^{\mathrm{RBF}} &= \exp\left(-\frac{\|z_i - z_j\|^2}{2D\sigma^2}\right), \\ A &= (K + \lambda I)^{-1} Y, \qquad \widehat{y}(z) = k(z)^\top A. \end{aligned} \tag{9}$$

Both use $\lambda = 0.001$; RBF bandwidth squared is the training-distance median, 2.21855. A fixed permutation of target changes supplies scrambled supervision. The edit is capped at twice the largest training-change norm, transformed back to native coordinates and written only into the declared window. Earlier attention entries and every convolution/recurrent state remain unchanged.

Policy application precedes the query. The function receives no corrected test state or expected answer, and its output hashes are committed before test donor processing. The same changed prefix is then forked into giver, recipient and color queries. This prevents the query-specific state construction that limited the earlier GPT-2 experiment. It remains a code-level information boundary, not a formally verified hostile-process sandbox.

Eight conditions compare unchanged state, learned linear and RBF corrections, scrambled supervision, direct role-row swapping, attention donor, complete donor and a fixed textual correction. The row swap does not compensate stored positional coding. The text condition adds a fixed instruction rather than the corrected story. Donor conditions have privileged corrected-prefix information. No method, prompt or threshold is selected from the held-out scores. Equation (9) describes ordinary supervised kernel prediction; this is not a full reproduction of ReFT or a claim that its proposed parameterization is optimal.

7.2 Results: state improvement without the requested relation

All 72 ordinary training answers and all 48 ordinary test answers are correct. The prospectively declared baseline threshold is at least 14 of 16 per query type. Scoring uses complete trimmed, case-folded generated answers, with exact-case outputs retained. Both legitimate fits have low training-state error: RMSE 0.014477 for linear and 0.003966 for RBF. The scrambled predictor's error against true changes is 1.882500. Shared preprocessing and all three fitted dual solutions store 5,363,726 float64 values, not merely the 20 coefficients of the earlier projected GPT-2 method.

Table 1 gives the full primary result. Neither learner produces one story with both roles correctly reversed, and neither passes a complete group. Their five successful role answers each arise from five same-name collapsed stories. The lower nonlinear training loss does not improve the relational endpoint. Both learners preserve the tested color attribute, demonstrating that one off-target success can coexist with failure of the intended transformation.

Table 1: Requested reversal and collateral preservation on 16 held-out prefixes. A complete group requires all six answers across opposite directions. The original condition's ordinary answers are correct; its role zeros indicate that it does not perform the requested reversal.

Condition	Giver /16	Recipient /16	Color /16	Groups /8
Original	0	0	16	0
Learned linear	1	4	16	0
Learned RBF	2	3	16	0
Scrambled supervision	0	0	11	0
Role-row swap	0	0	16	0
Attention donor	16	16	16	8
Complete donor	16	16	16	8
Fixed text correction	5	0	16	0

All seven non-original conditions change all 48 first-step full-logit hashes. Simple row swapping changes those internal outputs while leaving every original answer intact. Its descriptive mean raw partial-state error to the donor is 0.26553, smaller than unedited 0.37880, linear 0.36211 and RBF 0.35175. It still produces no requested role reversal. Figure 3 keeps target behavior, collateral preservation and complete groups separate.

The error comparison is descriptive and was not used to select an acceptance threshold. The norm cap never activates for the two correctly paired fits but activates in all 16 scrambled-supervision cases; the latter is not a realized-norm-matched label ablation. A single fixed textual correction is also unsuccessful here, which does not establish an inability to follow better prompts. No prompt search or larger instruction-following benchmark was performed.

The learned maps may fail because their parameterization, training sample, intervention support or preservation of the untouched recurrent groups is inadequate. Direct role-row exchange may be confounded by positional coding. The successful attention donor in this sample does not erase the earlier attention-only exception. Together the studies show that a supplied counterfactual state can work without making a learned approximation or reduced-state certificate sufficient.

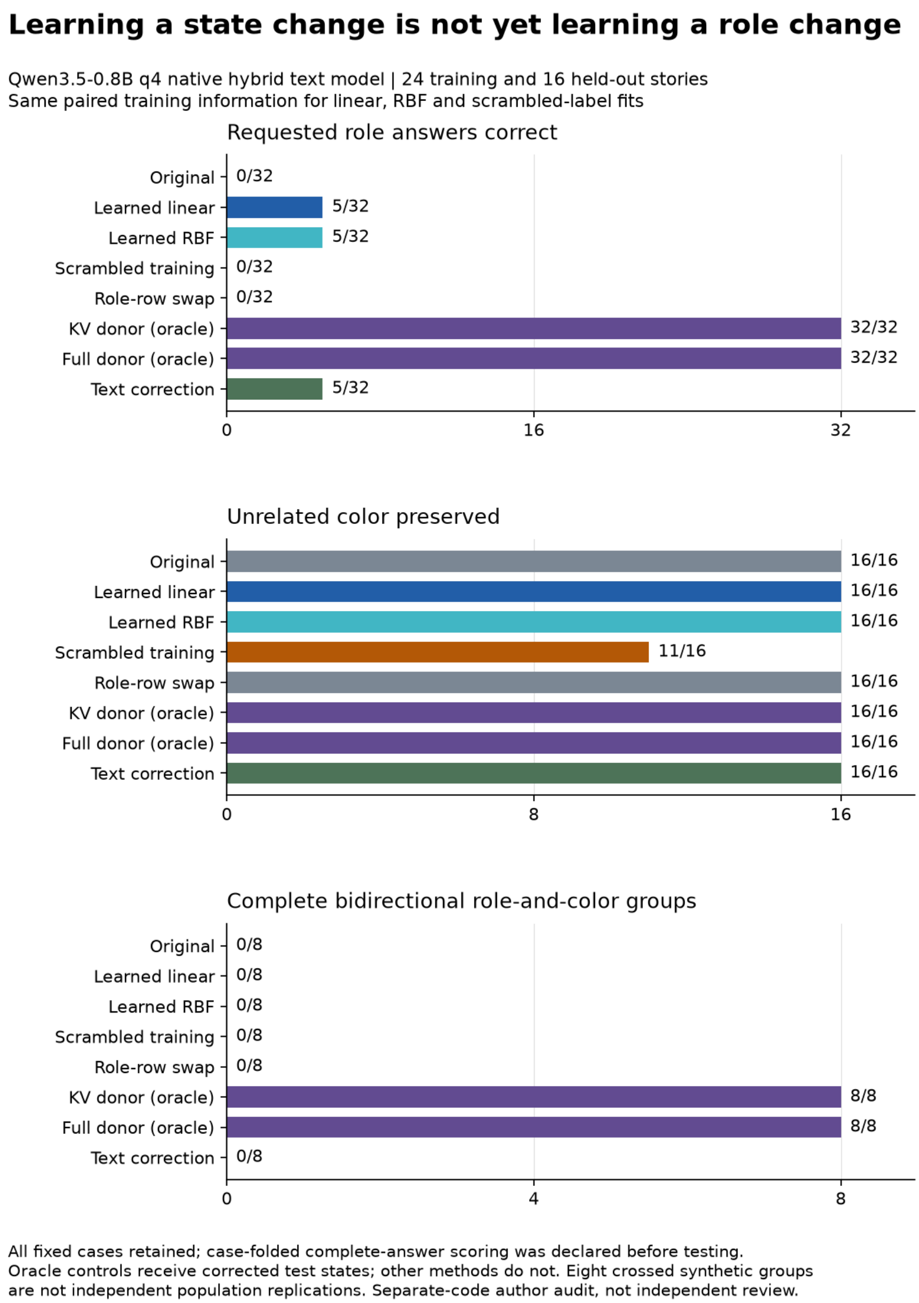

Figure 3: Frozen learned-correction results on the native hybrid decoder. The two learned maps preserve the tested color but fail relational reconstruction. Oracle success is a counterfactual-state result, not learned repair. Counts are shared with the Core Alignment companion.

7.3 Actual control sequence and reconstruction

One held-out prefix receives an authorized learned edit, authorized stop, forged restoration, authorized complete restoration and replayed old stop. Acceptance is true, true, false, true and false. Each accepted stage produces three fresh answers. The first accepted edit is behaviorally defective; stop retains that effect; complete restoration recovers the original answers from the known original state. This supplies an actual control sequence, not merely a hypothetical wrapper around saved scores.

The log does not retain raw signature envelopes for independent arbitrary-message cryptographic replay. It supports the scripted event and state-selection audit; prior fixture tests separately cover the monitor implementation. The key is public test material, not production security. The semantic and authority results must remain distinct.

Training, primary testing and workflow supply 72, 384 and nine answers respectively. Original collection uses 1,096 native calls. Twenty-two separate-code reconstructions repeat those calls and recover all 465 answer sequences, first-step full-logit hashes, training windows and learned test-state identities. They also check 6,144 state-support records and saved kernel equations. The largest measured original process commitment is below 0.70 GB, and original collection totals approximately 380 seconds in serial single-threaded jobs. This is separate-code self-review by the authoring process, not independent scientific replication or a new fresh full-pipeline replay.

8 Mathematical interpretation of the learned failure

8.1 Zero mean is not absence of structured information

The paired training permutation p is involutive. For raw states x_i, the desired changes satisfy

$$d_i = x_{p(i)} - x_i, \qquad \sum_i d_i = 0. \tag{10}$$

The equality follows because permutation leaves the sum of states unchanged. The least-squares constant predictor is consequently zero. This can hold while every individual pair has a nonzero and behaviorally relevant difference. A failure of mean-vector steering is therefore not evidence that the model lacks the relationship. The appropriate operation can be state-dependent.

8.2 A unique fit is not a semantic proof

For positive-semidefinite K, positive regularization implies

$$u^\top (K + \lambda I)u \geq \lambda \|u\|^2 > 0 \quad \text{for every } u \neq 0. \tag{11}$$

Thus Equation (9) has a unique coefficient matrix. The Gaussian kernel's positive semidefiniteness follows, on a finite set, by expanding its positive inner-product exponential into tensor-power kernels and multiplying by positive diagonal factors. Scaling the bandwidth and dimension preserves this property. The saved coefficient equations are numerically checked, with maximum residuals below 1.1×10^{-11} across the three fits. That checks the stated equations; it does not prove their learned outputs are useful.

8.3 The output boundary supplies a relevant sufficient condition

Let $L(t)$ denote logits from desired complete state t for one query. If token a is the unique winner with margin

$$m = L_a(t) - \max_{b \neq a} L_b(t) > 0, \tag{12}$$

then a proposed state s preserves that winner whenever

$$\|L(s) - L(t)\|_\infty < m/2. \tag{13}$$

Indeed, for any other token b,

$$L_a(s) - L_b(s) \geq m - 2\|L(s) - L(t)\|_\infty > 0. \tag{14}$$

For a complete answer, apply the same reasoning at each step conditional on the already matching token history, including the termination decision. A first-token contrast is insufficient. A justified bound on the complete downstream map could relate full-state distance to Equation (13), but the current study establishes no such Lipschitz bound or empirical margin certificate. Average training error on a partial attention window is neither a held-out full-state error bound nor a full-vocabulary output guarantee.

This is the specific certificate missing from a reconstruction-only interpretation. The point is not that all useful explanations must be globally exact. It is that the proposed explanation's claimed domain, retained state and behavioral tolerances must be connected. Equation (7) addresses sufficiency of a state projection; Equations (12)-(14) give one stronger output-level route under explicit assumptions. Neither follows automatically from a low probe loss.

8.4 Formal and executable scope

The shared package retains 31 compiled Lean statements, including seven original interpretation results, scope and passive-feedback results, and five paired-role metric statements. The relevant compiler and axiom audits contain no admitted proofs. Elementary components are axiom-free; the real-valued feedback arguments use standard logical axioms supplied through Lean's mathematical library. Nine projected finite cases and 120 role pairs connect specific executable scoring examples to the formal definitions.

The native hybrid-state sufficiency argument, ridge proof and output-margin derivation above are explicit mathematics, not newly compiled Lean results. None of the existing proof files formally refines the full Python, tokenizer, ONNX runtime and quantized model into Lean. The 84 passing application tests check declared construction and behavioral cases, not all possible implementations. Formal guarantees, numerical reproducibility and scientific interpretation therefore have separate acceptance boundaries.

9 Transformer relevance and the research program

The implementation map is concrete: GPT-2 supplies residual and attention-path interventions; Qwen supplies a contemporary hybrid decoder with additional retained state; DeepSeek supplies an architectural transfer target requiring different definitions. DeepSeek's compressed attention, altered residual transport and routed experts cannot be treated as ordinary GPT-2 cache rows with a new label [15]. No original DeepSeek experiment is reported. A distilled dense model does not reproduce the original mixture-of-experts architecture, and an API without validated internal access cannot establish the mechanistic claim tested here.

The results suggest three discriminating extensions. First, compare learned state operators with stronger representation-finetuning and causal-abstraction methods under matched training information and compute, with unrelated attributes beyond color. Second, test whether a successful operation survives novel queries, later generation, longer retained context and architectural changes, using baseline-competent development before freezing each holdout. Third, compare a model using internal feedback with an external controller having exactly the same information and action interface. Only an appropriate information difference or independently identified internal mechanism can support a stronger endogenous-control claim.

The present cases are synthetic and grammar-aligned. Training targets are supplied, spans are known, quantization is fixed, the models are small and the test families overlap by design. No human brain was read or stimulated, no clinical benefit was measured, and no experiment demonstrates an experience of selfhood. The underlying SAN account includes learning and recurrent biological consequence that these frozen-weight state interventions do not exhaust. Its artificial translation remains a program of specifying and testing candidate mechanisms, not a declaration that the current kernels embody the whole theory.

The source genealogy and research history also matter to interpretation. The donor leakage was repaired rather than hidden. The passive null and equal-information active tie remain part of the argument. The GPT-2 baseline floor is not converted into evidence of failed semantics. The initial Qwen capitalization failure remains distinct from the prospectively scored follow-up. These distinctions protect both the original proposal and the experimental conclusion from oversimplification.

10 Conclusion

Mechanistic interpretability becomes more demanding when an explanation is expected to support a useful correction. The receiving computation must implement the intended operation across the required questions and retained state, not merely expose a readable feature or change a logit. In the present native hybrid decoder, complete supplied counterfactual states succeed where the tested learned attention corrections do not. Equal attention state can coexist with different behavior, and lower partial-state error can coexist with no correct role reversal.

The recovered SAN question connects measurement, learned interpretation and returned consequence. The experiments make parts of that question operational while showing where the present implementations are inadequate. Their strongest contribution is the separation of decoding, causal effect, relational success, feedback learning and recoverability, with explicit mathematical and experimental obligations for connecting them. It is a bounded step toward causal self-regulation research, not evidence that autonomous or conscious self-regulation has already been achieved.

A Exact execution and evidence boundaries

Executed GPT-2 conversion, Xenova/gpt2: `bf2c7f02e0b826c60d03af341171bde20893da66`.

Qwen upstream model revision: `2fc06364715b967f1860aea9cf38778875588b17`.

Executed Qwen ONNX conversion: `fafab72d87a9e6be3925b38caf48286d2838f2d0`.

Qwen's selected quantized text dependencies total 666,439,935 bytes. The original model/runtime hashes, tokenizer configuration, generation template, stopping rules and individual predictions are retained in the companion reproducibility records. No claim of full-precision equivalence is inferred from the quantized result.

The Qwen experiment uses Python 3.12.14, NumPy 2.4.6 and isolated ONNX Runtime 1.29.0; the earlier GPT-2 adapter uses its separately pinned runtime. Numerical jobs use one thread and run serially. The learned fit uses 24 normalized 55,296-dimensional windows, fixed ridge regularization, training-only bandwidth selection and a prospective target permutation. Its held-out ordinary competence threshold, output normalization and group endpoints are fixed before testing. No generated answers are fed into other query branches.

The full earlier manuscripts and learned-study supplement remain available as research-history companions, with paragraph-level retention identities and an editorial destination map. They retain details condensed in the main scientific account, including early negative results, construction failures and original contextual qualifications. This preservation is an audit aid, not an assertion that an older draft is the current conclusion. Independent source, statistical, implementation and proof review remain outstanding.

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Declarations

This single-author manuscript was developed with AI-assisted research, coding and writing. Private clinical anecdotes are excluded. Original protocols, measured outputs, code, audits and formal logs are preserved. Separate code produced within the same authoring process constitutes self-review, not independent review. The Core Alignment companion shares the declared experiments without counting them as separate replications. Independent scientific, statistical, code and proof review and author readback remain outstanding. No human intervention, original DeepSeek execution or finding about phenomenal consciousness is reported.